

Finland, Sweden Warm Up to Data Centre District Heat Amid Lingering Sustainability Concerns

Category: International Data Center Taxation | Evidence ID: #124

Archived: 2026-10-04 20:16

- **Original URL:** <https://www.theenergymix.com/finland-sweden-warm-up-to-data-centre-district-heat-amid-lingering-sovereignty-concerns/>
 - **Research Category:** International Data Center Taxation
 - **Archival File:** 124_www_theenergymix_com_finland-sweden-warm-up-to-data-cen.pdf
 - **Extraction Status:** Full Web Transcript PDF
-

Source Content

In wintry Nordic countries like Finland and Sweden, some people are warming up to the idea of having data centres as neighbours, but concerns about “sustainable digitization” linger. Data centres in several countries with colder climates are channelling excess heat from their facilities through underground piping to warm nearby homes, Bloomberg News reports. Kai Mykkanene, mayor of Espoo, Finland’s second-largest city, said a new cluster of Microsoft data centres soon to be connected to the city’s district heating network will help eliminate local carbon emissions by 2030. “It’s the main solution to carbon neutrality for our energy,” Mykkanene said. Microsoft said the waste heat recovery will equal about 1% of the emissions reductions needed to meet Finland’s carbon neutrality target. According to Bloomberg, it will be the biggest heat recovery technology project anywhere in the world. When the data centre is built, the waste heat should be sufficient to warm about 100,000 homes though tens of thousands of kilometres of insulated pipes already in place under the city, Bloomberg reports. Hundreds of cities in North America have similar “steam districts,” and the infrastructure is common in Europe and Asia. The Nordic region’s electricity grid is dominated by renewable sources, and the frosty temperatures make it ideal for heat-generating servers amassed together in data centres. Along with some of the lowest electricity prices in the world, the region’s district heating systems have central generators and distribution systems already in place. Microsoft isn’t doing this out of the goodness of its heart. It is selling its waste heat for a price, and wouldn’t disclose the details of the Espoo contract to Bloomberg. The arrangement also raises questions about sovereignty and a back-up plan should Microsoft, an American-owned company, decide to shut down its facility. It’s one thing to depend on a data centre for cloud storage, quite another to depend on it for home heating. Google announced another multi-billion-euro data centre a little over a year ago for Muhos, a municipality in northern Finland, but then suddenly put it on hold in late October, 2025. The pause came after the Finnish government proposed to increase electricity taxes on data centres by removing an existing tax benefit. Another company, TLS Energy Oy, shelved its plans for a data centre over the same tax issue, said Bloomberg. Google’s parent company, Alphabet, said it was awaiting details on a tax benefit “under the terms of a separate model,” according to Yle, Finland’s national public service media company. That project had also planned an offsite heat recovery facility to serve about 80% of the neighbouring community, “free of charge.” Then there’s the massive amount of energy being consumed by these data centres, estimated by the International Energy Agency to be as much as 3% of total global electricity demand by 2030. Even in the Nordic countries cozying up to the waste heat, the electricity demands are “hugely controversial,” writes Bloomberg. “The data centre industry

speaks of this as if they are a power supplier because they are supplying heat, but they consume power we could use for other things,” Sofie Marhaug, a member of the Norwegian Parliament, told Bloomberg. The environmental impact assessment for the Microsoft facility “did not find a significant benefit to the climate,” said Reetta Suni, senior officer from the local authority. But another senior officer who assessed the Microsoft facility’s environmental impacts added that, “if we take it as a given that data centres need to exist, that we need them for our Zoom calls, then we need to find the best locations for them.” Veera Lyytikainen said that’s “where waste heat can be utilized and the data centre can be run on renewable energy.” Facing global backlash against their data centres, tech companies are ramping up efforts to associate with sustainability, and waste heat capture is one way to do that. A Bloomberg video referred to Finland as “redefining what it means to build a sustainable digital economy.” The United Nations Environment Programme advocates for what it calls “sustainable digitalization,” saying that although technology can help achieve environmental targets globally, it can also “exacerbate environmental impacts, deepen the digital divide, disrupt labour markets, and consolidate power among a select few over the majority.” Dr. Mél Hogan, associate professor of film and media at Queen’s University, has been writing about the impact of data centres, technology, and artificial intelligence (AI) for more than a decade. She told The Energy Mix that AI data centres “require so much energy to operate that by most calculations of scholars and technology critics, there is no way to ever recover your costs because of the energy demands alone.” “And so let’s say you had everyone on Chat GPT, paying \$200 a month for a subscription,” said Hogan. “That would not begin to recoup costs for AI.” That means the business model is “unsustainable because of the energy demands and then because of the energy demands, it is environmentally unsustainable.” She questioned the concept of sustainability in our current environment and the trade-offs people are willing to make when it comes to AI data centres. “There are arguments towards sustainability that are legitimate, but it is a bigger question about what is being sustained,” Hogan said. “What’s complicated about the concept is that we can’t sustain the present moment because we’re falling off a cliff in terms of global warming.” That has her wondering about the “future that we’re being pitched,” and how many actually believe that AI is inevitable. “We’re told it’s here to stay that we just have to be patient and the big promise is that finally something will come to fruition that will really pay off,” Hogan told The Mix. “But the harms are here and there are a lot of issues we need to address now, regardless of whatever sort of imagined future potential there might be.”